

The first operating system

To ensure that the precious computers weren't wasted, companies and educational institutions needed a system which ensured that programs are always running on it.

Believe it or not, the first operating system was human—full-time computer operators who were hired to make sure that the computer would never be idle. Programmers would come to them with their programs on a punch-card and tell them to execute it. The operator would run the program, record the output, and give it to the programmer who would return after a nice snack.

The first *real* operating systems

Soon enough, programs were written that would perform the tasks of the computer operator—they would run a program, free up the resources used by it, and move on to run the next program. These were the first computer operating systems.

Every programmer used to write his or her own code to generate an output for their program, and the makers of the OS wanted to relieve them of this as well. This led to the introduction of the *device driver*—programs that could already talk to hardware. So programmers now just needed to write code to talk to the driver, rather than write their own drivers. It was also decided to give programmers a set of *library functions*—basic functionality that nearly all programmers would use. The management capabilities, device drivers and library functions together made up the structure of the OS—a structure it has even today.